[2 marks]

c) State the correct value of the corrupted bit: **It should be 1 instead of 0. [1 mark]**

7. Check Digits & ARQ: (5 Marks)

a) Differentiate between a *Transcription Error* and a *Transposition Error*: (2 Marks)

Transcription: Mistyping a single digit (e.g. typing 3 instead of 8).

Transposition: Swapping the position of two adjacent digits (e.g. typing 78 instead of 87). [2 marks]

b) In an ARQ protocol, what happens if the sender does not receive an ACK before the timeout timer expires? (3 Marks)

Model Answer: The timer times out, and the sending computer automatically re-transmits the same packet. This cycle continues until a valid ACK is returned or the retry threshold limit is reached. [3 marks]